Quham Adefila

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Education

Imperial College London, Computing (MEng): 1st Class Oct 2020 – Oct 2024
Modules: Deep Learning, NLP, RL, Computer Vision, Distributed ledgers, Finance, Computing Practical, Linear Algebra etc.

Brampton Manor Academy Sep 2018 – June 2020
A-Levels: Maths (**A***), Further Maths (**A***), Computer Science (**A***), Physics (**A***)

Work Experience

Rokos Capital Management, Technology Analyst Sep 2025-Current

- Improved decision speed across credit desk with real-time dashboard monitoring bond gainers/decliners,
- Automated data validation with workflows flagging and correcting anomalies, saving **several hours** investigation.
- Proposed and led implementation of equipment request system, streamlining turnaround from **weeks to minutes**.

Scope, Founding Engineer (Contract) May 2025-Jun 2025

- Led RBI project representing **50%** of company ARR (**~£300k**)
- Revamped interface ongoing for 9 months, prioritizing inspector needs and enabling **first successful** usage.
- Developed system to automate client onboarding, reducing onboarding time from **weeks to minutes**.

Maia Cares, Founder Mar 2025-Sep2025

- Gained deep understanding of social workers' day-to-day challenges through direct engagement and observation
- Built MVP used by 80+ social worker and nurses, based on validated problem area, tested in target network.
- Connected with Heads of Service and Chief Social Worker at the Department for Education discussing pilots.

Google Deepmind, Research Engineer Intern – Vision Jul 2024 – Aug 2024

- **Predicting** future typhoon frames and pressures using **video models** such as DINOv2, VideoPrism, V-JEPA
- Handling **pipeline** from dataset onboarding to **finetuning** and using high capacity output layers using **TensorFlow** and **JAX**.
- Repurposing models with RGB input to handle **infrared** images achieving performance **30%** more accurate than benchmark.
- **Learned to:** Effectively use GDM internal tools, implement a full ML pipeline in industry, finetune **spatiotemporal** models.

JP Morgan, Software Engineering Intern - Equities, Prime Finance Apr 2023 - Sep 2023

- Implemented multiple APIs for **data sourcing** and **file handling**, in Python for data visualisation application used by traders
- Optimised write speed by **718% KDB database** usage **migrating data API** and implementing **port selection algorithm**.
- Employed **clustering algorithms** to analyse extensive **datasets with millions** of entries, using Numpy and Pandas
- Streamlined and enhanced script **automation processes** for increased simplicity and efficiency for entire equities division
- Identified and rectified **data inconsistencies**, leading to improved data handling efficiency and a better maintained database
- **Learned to**: manage legacy software, handle large datasets, optimise code maintaining clarity, work in a global team

Improbable, Software Engineer Intern – Defence, Modelling Jun 2022 - Sep 2022

- Integrated civilian model into an **operational decision-making simulation** tool for military use
- Improved setup for future integrations resulting in **a week's timesaving per team member.**
- Streamlined the process to introduce components to the civilian model, enhancing model **adaptability**
- Expanded **probabilistic Python models**, to **simulate** power outage and bombing effects on civilian migration and casualties, aiding **strategic planning** and **risk assessment,** deployed with **Kubernetes**
- **Learned to:** collaborate in an **agile** environment (scrum), enhance interconnected Python models

Inspire Maths Tuition, Founder Sep 2018-Sep 2024

- Tutored math and science to ages 7-19 with **100%** of students exceeding exam targets and **lead** a team of **10 tutors**.
- Developed a **website** to facilitate the **transition** to online lessons during the pandemic also organising free lessons rated 4.6/5
- **Learned:** web development (JavaScript), leadership, marketing, communicating and explaining clearly

Projects

Multi-asset backed Stablecoin, with verifiable reserves Jan 2024 – Jun 2024

- Creating a **stablecoin** to enhance **financial inclusion** with the technical components of cryptocurrencies abstracted.
- **Optimising** for a **multi-asset** peg across traditional and alternative assets to achieve a return **outpacing inflation**.
- Leveraging **garbled circuits** in TLS allowing **independently verifiable reserves** using TLSNotary and **Rust.**
- **ReactJs** frontend with **Python Flask** backend and **MongoDB** database, with UI designed for **convenience** through feedback.
- Learned to: deploy **Solidity** smart contract on **Polygon**, integrate separate technical components producing a effective Dapp

3D snooker reconstruction using Computer Vision, Python Sep 2022 – Dec 2022

- Detected and tracked snooker balls from single camera footage handling various camera angles
- Tested Yolo algorithm, refining using OpenCV, and implementing tailored ball **tracking algorithm**
- Analysed snooker gameplay: identifying potted balls, speed tracking, **modelling** alternate shots
- Learned to: use **computer vision algorithms/libraries**, and develop and refine custom algorithms

Short Experience

- **Gresearch (2022):** Quant challenges and workshops on Programming, Finance, Stats and ML. **1st** place in trading challenge.
- **Jane Street (2021):** Insight into trading and software engineering while solving math and programming challenges.

Skills and Interests

- **Programming:** Python, C, Java, Haskell, **Other:** Kotlin, Scala, SQL, JavaScript, Solidity, Rust
- **University Societies:** Fintech, Blockchain, Basketball, **Other Interests:** Table Tennis, Gym, Teaching